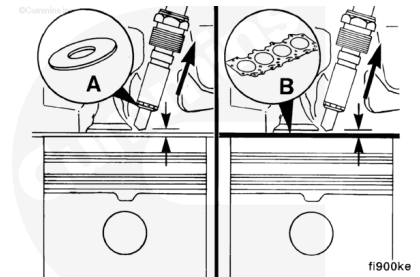


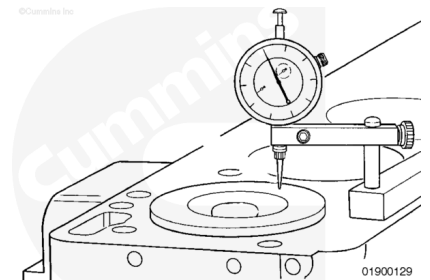
Trainings ▾

General Information

Injector protrusion can affect power from the engine. In addition to a single sealing washer (A) on the injector, the thickness (B) of the head gasket controls injector protrusion.

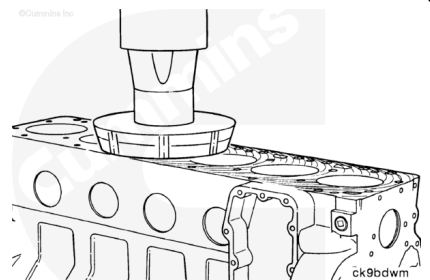


Piston protrusion can affect startability and engine smoke output. The cylinder head gasket helps to maintain correct piston protrusion.

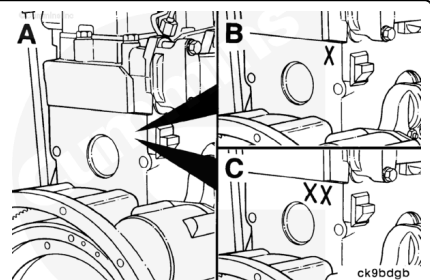


For B3.9, B4.5, and B5.9 series engines, oversize cylinder head gaskets are available for resurfacing of the cylinder head and cylinder block combustion decks to maintain correct:

1. Injector protrusion
2. Piston protrusion.



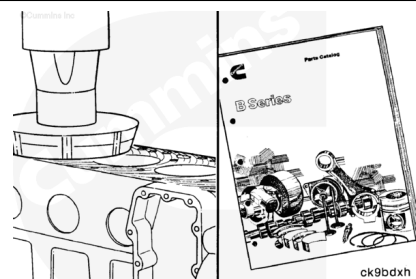
After machining, the cylinder is stamped in the upper rear right hand corner surface of the cylinder block (shown). The cylinder head is stamped on the rear of the cylinder head above where the cylinder block would be stamped (**not** shown).



Machining	Mark
A - Standard	None
B - 0.25 mm [0.010 in] machined for first oversize gasket.	X

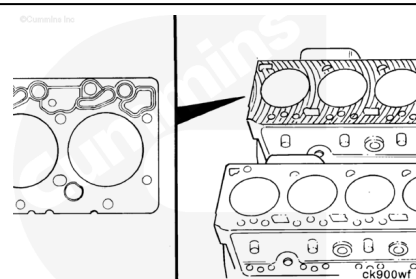
Machining	Mark
C - 0.25 mm [0.010 in] machined (0.50 mm [0.020 in] total) for second oversize gasket.	XX

After determining the amount of machining that has been performed, see parts catalog for the proper oversize head gasket.



For B4.5 RGT series engines, the combustion deck of the block can only be resurfaced if, after resurface, the correct piston protrusion can be achieved. Refer to Procedure 001-054 (/qs3/pubsys2/xml/en/procedures/40/40-001-054-tr.html).

No specific head gasket with an increased thickness is available for combustion deck resurfacing. If the combustion deck can **not** be resurfaced such that the correct piston can be achieved, the cylinder block **must** be replaced.

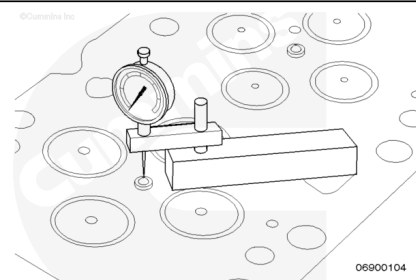


For B4.5 RGT series engines, the combustion deck of the cylinder head can only be resurfaced if both of the two following criteria are met:

1. Valve depth is kept in specification. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html) for specifications
2. Injector protrusion is kept in specification. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html) for specifications.

If valve depth and injector protrusion specifications can be maintained, the cylinder head can be resurfaced. If the specifications can **not** be maintained, the cylinder head **must** be replaced.

No specific head gasket with an increased thickness is available for cylinder head combustion deck resurfacing. If the combustion deck can **not** be resurfaced such that the valve depth and injector protrusion can be reached, the cylinder head **must** be replaced.

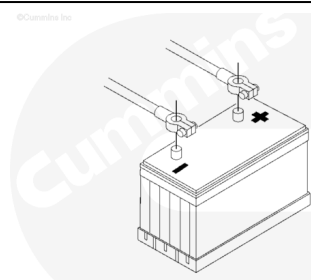


Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries.

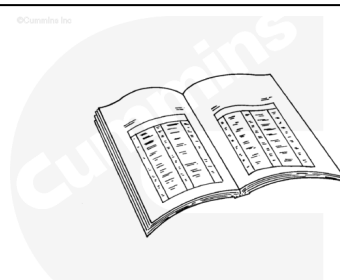


13900050

⚠ WARNING ⚠

This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

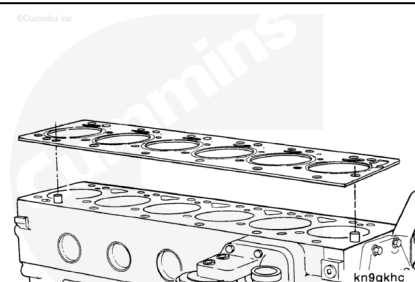
- Remove the cylinder head. Refer to Procedure 002-004 (</qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html>).



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Remove

Remove the head gasket.

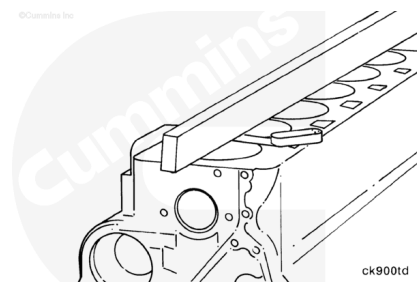


kn9gkhc

Clean and Inspect for Reuse

Note : Cylinder head gaskets can not be reused. A new cylinder head gasket must be used.

Clean the cylinder block and cylinder head combustion deck. Inspect and measure the cylinder block and cylinder head combustion deck flatness. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html).



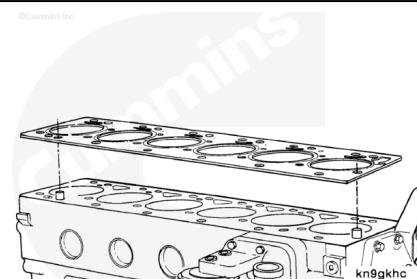
Install

⚠ CAUTION ⚠

Be sure the gasket is correctly aligned with holes in the cylinder block. If the gasket is not correctly engine damage can result.

Install the new head gasket.

Note : Never reuse the old head gasket. Always use a new head gasket to prevent leakage.

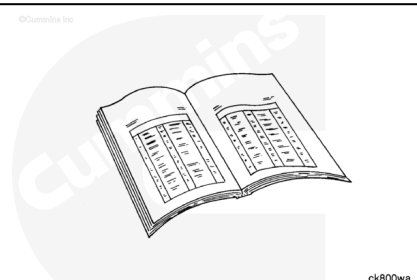


Finishing Steps

⚠ WARNING ⚠

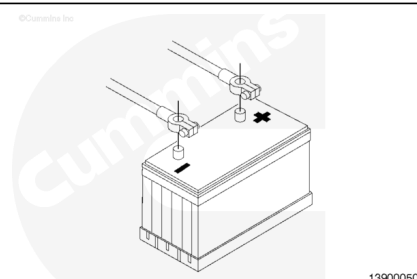
This assembly weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this assembly.

- Install the cylinder head. Refer to Procedure 002-004 (/qs3/pubsys2/xml/en/procedures/40/40-002-004-tr.html).



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,



remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Connect the batteries.

Last Modified: 24-Apr-2006
